SOME CHANGES OF MATH ANXIETY GROUPS BASED ON TWO MEASUREMENTS, MASS & EEG

Byeongguk Ryoo and Sang Sook Choi-Koh

Dankook University

This article investigated how mathematics anxiety (MA) of Korean middle school students could be reduced by comparing analytically their cognitive neuroscience and questionnaire results. We developed a three-hour Complex Treatment Program (CTP) on quadratic functions for the study. In the summer of 2016, we collected data of the pre and post MA questionnaires by Mathematics Anxiety Scale for Students (MASS), the percent of correct answers (PCA) & reaction time (RT) by E-prime program, and also brain-imaging data of the event related potentials (ERP) by Electroencephalo-graph (EEG) using computer-based functional F-G model. The result indicated the CTP to be effective with the group with higher math anxiety and the group with higher achievement respectively. The MASS result was verified with the better performance of PCA on type G, which was measured by E-prime program. Some interesting patterns were revealed on brain-imaging data by EEG, indicating more mental activities with the MA groups.

INTRODUCTION

Looking back at the Programme for International Student Assessment (PISA) from 1995 to 2012, Korean students ranked in the highest level of achievement in mathematics but consistently marked a very low score in affective domains like self-efficacy and interest (Ko & Yi, 2012). This leads us to conclude that students who are high-performers in mathematics could potentially have low achievement levels in the affective domains of mathematical learning, resulting in mathematics anxiety (MA) which is displayed as the uneasy feeling about the subject of mathematics. While many previous studies about MA were centered on the development of questionnaires to diagnose students' MA, recent studies have been focused on devising a treatment to abate it. For example, researchers applied the Note-taking Model like Divided-page Method (Tobias, 1987; Eun & Lee, 1994), and the Mathematics-friendly Program (Yoon & Jeon, 2010) to students who were feeling mathematics anxiety. However, these studies focused mainly on problem solving or simple algorithms and not on a specific domain of mathematics, e.g. function. In order to reduce MA in the students' functional understanding of mathematics, we developed a complex treatment program that combined a mathematical treatment and a psychological treatment, once used by Yoon et al. (2010). Moreover, we collected data to grasp the effects of the program more closely by measuring the students' brain waves which is the difference of this study from previous ones.

2018. In E. Bergqvist, M. Österholm, C. Granberg, & L. Sumpter (Eds.). *Proceedings of the 42nd Conference of the International Group for the Psychology of Mathematics Education* (Vol. 4, pp. 83-90). Umeå, Sweden: PME.

The purpose of this study was to investigate how to reduce the MA of eighth-grade students after they completed the complex treatment program (CTP), which was a mixture of psychological and mathematical treatments to help students psychologically and mathematically. The research questions were set up as follows:

1. How did the student groups respond to the MASS before and after the complex treatment program?
2. What were the differences of the groups in the percent of correct answers (PCA) and reaction time (RT) gained by E-prime program while getting the brain imaging data through ERP?
3. What were the differences in the brain imaging data of the groups on the functional tasks through ERP?

THEORETICAL BACKGROUND

Mathematics Anxiety (MA)

In the previous research, various researchers developed differing definitions of MA(e.g. Fennema & Sherman, 1976). Since Byrd's (1983) comprehensive definition, several researchers have defined MA differently and mostly used the definition formerly described in the literature. Various definitions, however, were similar in the context of mathematical learning. In this study, we defined MA as "anxiety felt by individuals when they solve mathematical problems".

Brain Research on Students' Understanding of Functions

Akkoc and Tall (2002) suggested the complexity of the function concept arises from its many representations, e.g. graphs, equations, and tables. In our study, we decided to focus on the translation between graph and equation representations—a particularly important concept in understanding functions (Williams, 1998). As such, our study resembles two previous neuroscience studies on students' understanding of functions (Thomas Wilson, Corballis, Lim, & Yoon, 2010; Waisman, Leiken, Shaul, & Leiken, 2014) though neither of them considered the effects of MA.

Thomas et al. (2010) classified function tasks into four formats: graph-to-graph, equation-to-equation, graph-to-equation and equation-to-graph. They used these formats for both linear and quadratic functions, generating a total of eight kinds of tasks for the ten students in their study—ten college students in New Zealand. In addition to measuring RT and PCA, they measured neural activity within the neocortex of the brain through use of functional magnetic resonance imaging (fMRI). Essentially, fMRI measures neural activity by way of blood flow to areas within the brain.

METHOD

Participants

25 students in the 8th grade from middle school in Yongin-city, Korea participated in this study. The students had never participated in a program for the purpose of reducing MA.

Procedure

The data was collected in July, 2016 in which the subjects were available right before the summer vacation started in Korea. In this study, the pre and post-questionnaires by MASS were used to measure students' MA. In the same way as Sheffield & Hunt (2006), the participants were classified as high MA (HAX) group or low MA (LAX) group based on the result of the pre-questionnaire. Also, according to the result of the recent final exam of their school, they were grouped into either the higher achievement (HAC) group or lower achievement (LAC) group. The treatment program of the three-lesson units was applied to these students. The students reviewed the problems on their own, while working with the experimenter. To find the effectiveness of the treatment program, we investigated the degree of MA using the same questionnaire as the pre-questionnaire and measured the students' brain waves to see how they reacted to stimuli of functional tasks while they solved them on the monitor provided.

Instruments

Questionnaire for MA. The Mathematics Anxiety Scale for Students (MASS) revised by Ko & Yi (2012) was used to measure students' MA. This test scale was chosen because it was more appropriate to the middle school students since their MASS was developed for the secondary school students including middle school students with 65 question items which resulted in the internal consistency reliability, the Cronbach's α= .976 as reported in table 3. Each question was answered on one of 1 to 5 point Likert scale.

Complex Treatment Program (CTP). The complex treatment program in three units consisted of the mixture of a mathematical treatment and a psychological treatment. The mathematical treatment that was composed of four kinds of problems provided students a chance to reflect the nature of quadratic function. Firstly, the diagnosis problem was constructed as the basic concepts learned in the previous lesson. Secondly, a lower level (basic) problem was constructed with a basic concept of the quadratic function. Thirdly, the treatment was constructed to help students to understand the type G. Lastly, a higher level (challenging) problem was constructed to challenge advanced level students with a higher level of difficulty. Also, four to six mathematical-hint cards reflecting Vygotsky's scaffolding theory in which they could be assisted by peers or a teacher whenever needed were prepared for each problem.

The psychological treatment which was the other part of CTP was constructed by revising the brain integration in education (BIE) program used by Kim (2010) to reduce the students' anxiety factors shown in the pre-questionnaire. Kim (2010)

developed the brain integration in education (BIE) program consisting of three stages: understanding of brain, integration of brain, and application of brain, to improve self-directed learning ability of middle school students.

F-G Model of Functional Translation. Thomas et al. (2010) presented the functional tasks in relation to graphs and equations as stimuli and observed 10 students' brain waves. By actively using it, the tasks developed in this study were composed of 10 kinds of basic quadratic functions with $a = \{-1,\ 1\}$, $p = 0$, $q = \{-2, -1,\ 0, 1, 2\}$ in a algebraic formula, $y = a(x-p)^2 + q$. In other words, the subject was able to grasp the elements such as shape, intercept, vertex, etc. of a quadratic function.

Two hundred items were developed which asked students to decide whether graphs and equations matched. E-prime program randomly decided an item and presented it the subject. Two types of cross-format functional tasks were translating from equation-to-graph (type F) and translating from graph-to-equation (type G) using the ten quadratics functions. The students solved 20 problems as presented randomly for each type.

Brain-Imaging Technique. For this study, Electroencephalograph (EEG) was selected. This decision was based on budget, mobility, accessibility, safety, functionality, and practicality. In detail, we selected Event Related Potential (ERP) among the EEG's techniques. ERP is a method of averaging EEG activity from time domain analysis and multiple-stimulation in an EEG. It usually appears within 50-500ms after stimulation: P50, & N100 which reflect the stimulus detection, and cognitive-related factors, P300, & N400 which appear later and mainly indicate stimulation.

RESULTS

Pre and Post Questionnaires

The mean value of the pretest for 25 students was 2.63. The students who were higher or lower than 2.63, were classified as either HAX (High Mathematics Anxiety) or LAX (Low Mathematics Anxiety) respectively. Also, students who were higher or lower than the final exams' mean value, were grouped as either HAC (High Mathematical Achievement) or LAC (Low Mathematical Achievement) according to the recent final exam administered by their school, to see relationship between cognitive and affective aspects in detail.

The following Table 1 compares the results of the pre-questionnaire by groups with the results of the post-questionnaire after the program was implemented. This comparison in two dependent sample t-tests was to see the effect of the complex treatment program.

Group		Mean	SD	N	t value
HAX	pre	3.41	0.68	11	2.41*
	post	2.94	1.01	11	
LAX	pre	2.02	0.52	14	1.21
	post	1.82	0.75	14	
HAC	pre	2.23	0.72	15	2.57*
	post	1.88	0.72	15	
LAC	pre	3.23	0.86	10	1.08
	post	2.96	1.11	10	
All	pre	2.63	0.91	25	2.51*
	post	2.31	1.03	25	

*p < .05

Table 1: Two dependent samples t-test results for the effect of the CTP in groups

The Percent of Correct Answer (PCA) and Reaction Time (RT)

In order to measure the responses of the students who participated in the research, we developed the functional types for the E-prime program along with EEG measurement, which provided us the percent of correct answers (PCA), the reaction time (RT) of the subjects. Before solving the F-G Model, the students solved arithmetical tasks (type A) consisting of simple addition problems to help activate their brain. This is called the dual-task composition in which the students worked on two phases: first, the phase for arithmetical tasks; second, the phase of functional tasks to capture more proper activation of brain. However, the result of type A was not included in this paper. PCAs and RTs of four groups for each type of tasks are as follows.

Group	Type F		Type G	
	PCA	RT	PCA	RT
HAX	72.7%	2.05 sec	86.4%	1.8 sec
LAX	83.6%	1.78 sec	84.2%	1.41 sec
HAC	88.5%	1.62 sec	89.7%	1.26 sec
LAC	66.7%	2.25 sec	79.5%	1.98 sec

Table 2: PCA and RT by group for the type F and G

EEG

In this study, we analyzed P300 wave in Event-Related Potential (ERP) to measure brain waves, in addition to PCAs & RTs by E-prime program. The P300 wave is a positive wave that reaches the apex about 300 ms after stimulus presentation starts in ERP. The results of measuring brain waves while solving types F and G through EEG are shown in Fig. 1 which compares the groups for the types. In this paper, we just displayed the imaging data for groups because of spatial constraint. In the conference site, we will display all data we gained.

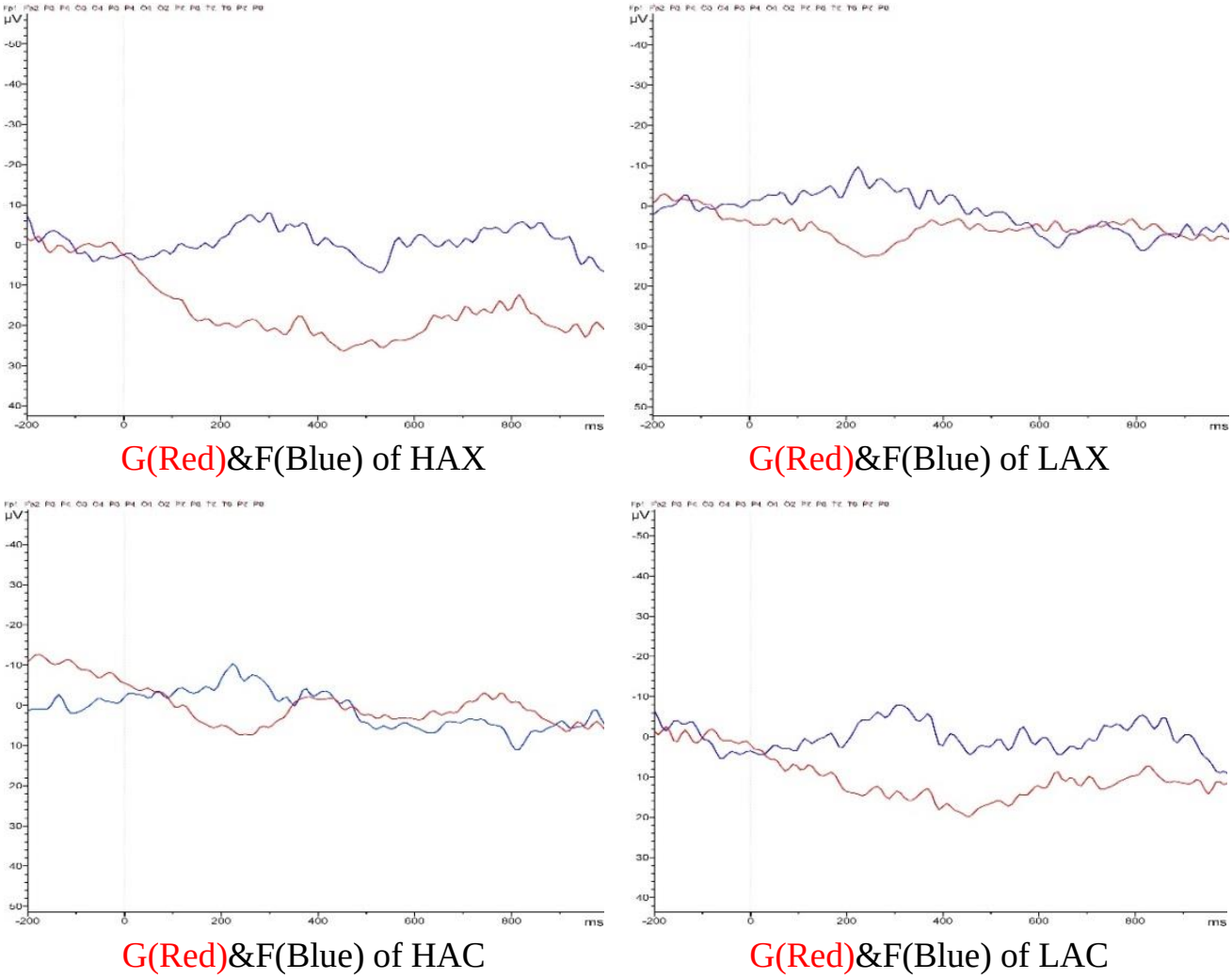

Figure 1: EEG by a task for the group

CONCLUSIONS

First, the test results by MASS showed that our program overall was effective for the students to reduce MA. In detail, MA of group HAX and group HAC had significantly been lowered. In fact, because our program was composed of two important parts that help students with psychological and mathematical aspects, this outcome on Tab. 1 made sense to us.

Second, after completing our program, all the groups of students solved type G more accurately and quickly than type F. That is, the CTP we had developed to help the students not only to reduce MA, but also to get over the difficulty on type G for better functional understanding played an important role in achieving the purpose of the study. Interestingly, the result by E-prime that HAX and HAC exceeded the counterpart on type G in Table 2 seemed to verify the result we got by MASS outcomes.

Third, to examine the precise effect of the program, we observed closely at the brain waves by EEG while the students were solving the F-G Model. As mentioned above, although the students solved type G more quickly and accurately, compared to type F, we noticed that the students had more brain wave activity with type G in brain (see Fig. 3). Also, the MA groups presented a bigger gap between two types, which meant more activities on type G in brain. If we had not measured EEG in addition to the questionnaire of MA, we might not have found this kind of difference of type G on which students were asked to decode the graph image first, and then to match it to a corresponding equation of function. Therefore, the MA aspect influences more on students' performance.

Today, the advent of cutting edge-handy devices for measuring brain waves makes it possible to apply them for educational purposes like reducing MA. The importance of mathematics is increasing day by day, but there are many students, who have a sense of discomfort with mathematics. For the sake of national core competence, we should solve the MA issue and develop new methodologies in mathematics education.

References

Akkoç, H. & Tall, D. (2002). The simplicity, complexity and complication of the function concept. In A. Cockburn and E. Nardi(Eds.), *Proceedings 26th Annual Conference of the International Group for the Psychology of Mathematics Education, 2*(pp. 2-025). Norwich, UK.

Byrd, P.G. (1983). A Descriptive Study of Mathematics Anxiety: Its Nature and Antecedents. *Unpublished Doctoral Dissertation*, India University.

Eun, S.J. & Lee, J.H. (1994). A Study on an Effective Teaching Method for the 7th Grade Students with Math Anxiety. *Mathematics Education*, *33*(2), 285-296.

Fennema, E., & Sherman, J. A. (1976). Fennema-Sherman mathematics attitudes scales: Instruments designed to measure attitudes toward the learning of mathematics by females and males. *Journal for research in Mathematics Education*, 7(5), 324-326.

Kim, E.J. (2010). A Study on Development of a Brain Education Program to Improve Self-Directed Learning Ability. *Korean Journal of Youth Studies*, *17*(10), 161-188.

Ko, H.K. & Yi, H.S. (2012). Difference in Mathematics Anxiety of Middle and High School Students per Factor according to Background Variables, *Journal of the Korean School Mathematics Society*, *15*(3), 487-509.

Thomas, M.O., Wilson, A.J., Corballis, M.C., Lim, V.K. & Yoon, C. (2010). Evidence from Cognitive Neuroscience for the Role of Graphical and Algebraic Representations in Understanding Function. *ZDM*, *42*(6), 607-619.

Waisman, I., Leikin, M., Shaul, S. & Leikin, R. (2014). Brain activity associated with translation between graphical and symbolic representations of functions in generally gifted and excelling in mathematics adolescents. *International Journal of Science and Mathematics Education*, *12*(3), 669-696.

Yoon, R. & Jeon, I.H. (2010). The Development of a Math Friendly Activity Program for the Alleviation of Mathematics Anxiety. *Journal of Elementary Education in Korea*, *14*(3), 583-613.